# Short Answer Test

**31/08/2026** **P.Sai Ganesh Reddy**

**CSA0311** **192525111**

Q1 **Does Floyd–Warshall compute single-source or all-pairs shortest paths?** 2 marks

Does Floyd–Warshall compute single-source or all-pairs shortest paths?

Your Answer

Floyd–Warshall computes all-pairs shortest paths. It finds the shortest distances between every pair of vertices in a weighted graph, rather than just from a single source vertex to all others (which is what algorithms like Dijkstra's or Bellman-Ford do).

Save Answer Next →

Q2 **How many edges must a Minimum Spanning Tree of a graph with $V$ vertices have?** 2 marks

How many edges must a Minimum Spanning Tree of a graph with $V$ vertices have?

Your Answer

A Minimum Spanning Tree (MST) of a graph with V vertices must have exactly V−1 edges.

Save Answer Next →

Q3 **In a DAG, which node will always have an in-degree of zero in a topological ordering?** 2 marks

In a DAG, which node will always have an in-degree of zero in a topological ordering?

Your Answer

In a DAG, the source node (the node with no incoming edges, i.e., no predecessors) will always have an in-degree of zero in a topological ordering.

Save Answer Next →

Q4 **Differentiate between directed and undirected graphs.** 2 marks

Differentiate between directed and undirected graphs.

Your Answer

In a directed graph, edges have a specific direction, meaning each edge goes from one vertex to another in a single direction (like a one-way street). This is represented using ordered pairs, e.g., (A, B) means an edge from A to B only, and it does not imply an edge from B to A.
In an undirected graph, edges have no direction, meaning the connection between two vertices works both ways (like a two-way street). This is represented using unordered pairs, e.g., (A, B) means there is a connection between A and B, and it is the same as (B, A).

Save Answer Next →

Q5 **Define a connected graph.** 2 marks

Define a connected graph.

Your Answer

A connected graph is a graph in which there exists a path between every pair of vertices.

Save Answer